(1) $$y = \Sigma_i \; \boldsymbol{p_i} \cdot W_2 \, \varphi(z_i)$$

(2) $$y = \Sigma_i \, W_2 \, (\boldsymbol{p_i} \cdot \varphi(z_i))$$

valid when W_2 is linear, no bias

$$\boldsymbol{\partial L / \partial p_i} = \langle \, \varphi(z_i), \, \partial L / \partial a_i \, \rangle$$

ϕ(z) rebuilt from z in the kernel

z is saved for SwiGLU anyway